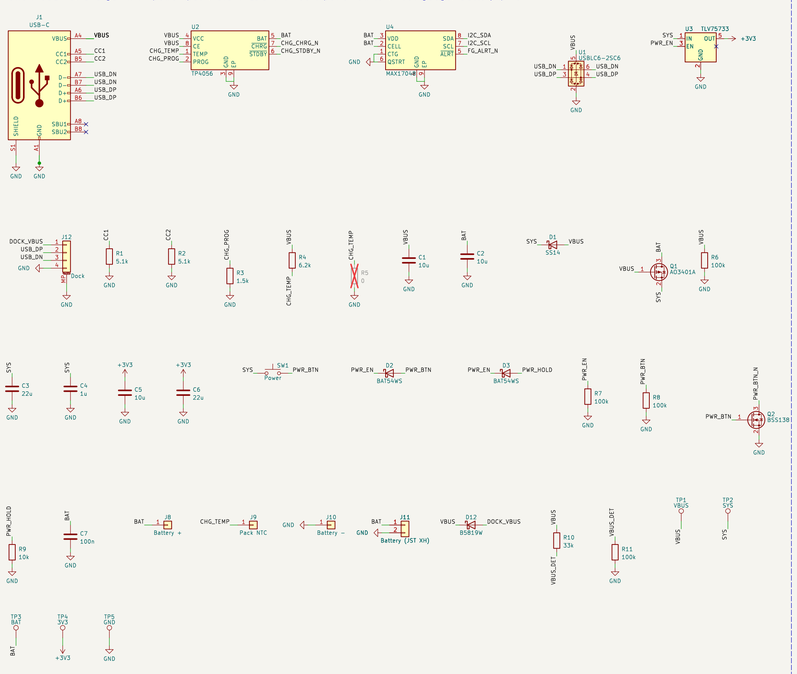
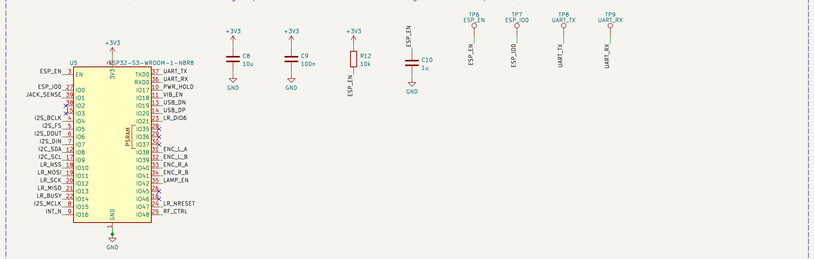
OpenTheatreComms beltpack — main PCB

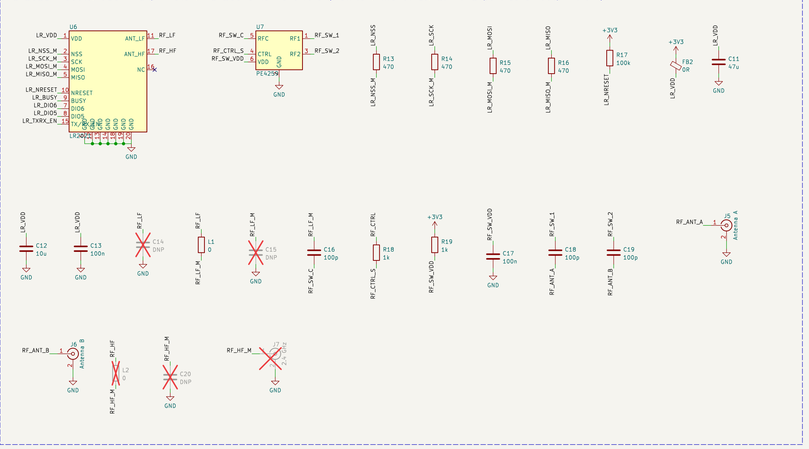
Power: USB-C, charger with power path, 3.3 V LDO with soft power switch, fuel gauge, battery pins



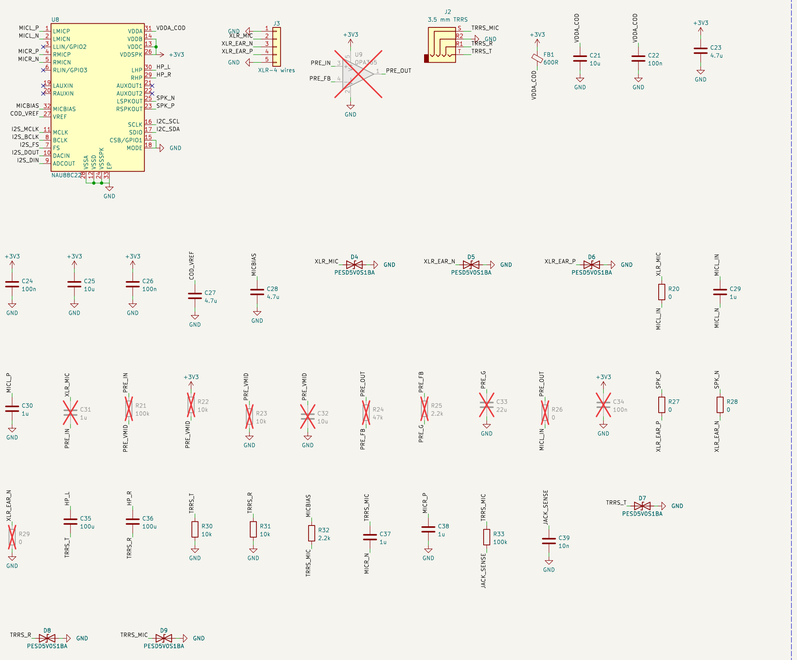
ESP32-S3-WROOM-1-N8R8: radio timing, Opus, UI. Native USB for flashing and serial; pads for EN, IO0 and UART0



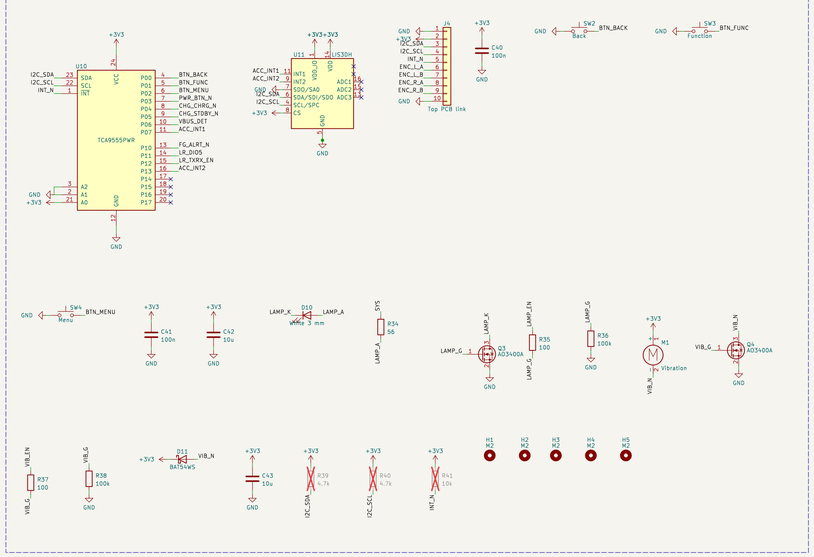
LR2021 module, 915 MHz pi match, PE4259 antenna diversity switch, two U.FLs; 2.4 GHz port as a footprint only



NAU88C22 codec: XLR-4 dynamic headset (BTL earpiece), 3.5 mm TRRS headset (stereo, electret), optional mic preamp slot



IO expander (0x21), front buttons, LIS3DH accelerometer, flashlight, vibration motor, link to the top PCB, mounting holes



PWR_FLAG PWR_FLAG +3V3 GND

Power, ESP32-S3, LR2021 radio and RF, codec and headsets, IO. Generated from pcb_main/design.py
OpenTheatreComms
Sheet: /
File: main_pcb.kicad_sch
Title: Beltpack main PCB
Size: A1 | Date: 2026-10-26 | Rev: 0.1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1